Bayesian Time-Varying Multi-Seasonal ARMA Models

Stable and Scalable Inference for Nonstationary Time Series

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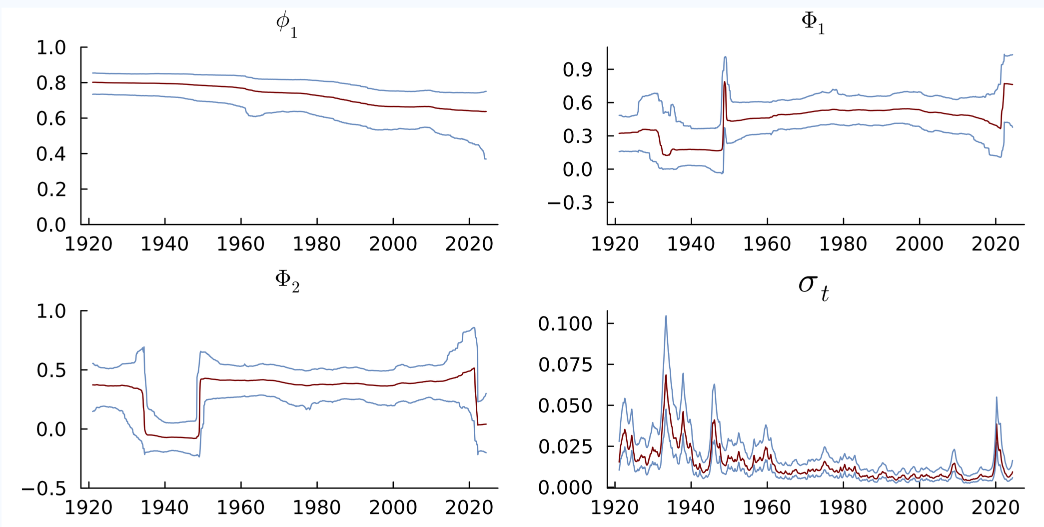
1. Why TV-SARMA Models?

- Smooth evolution and abrupt shifts
- Multiple seasonalities (daily, weekly, annual)
- Adaptive models



TV-SAR$(1,2)_{12}$: posterior median and 95% credible bands.

2. Modeling framework/contributions

- Bayesian TV-SARMA
- Multiple seasonalities
- Nonlinear state-space inference
- Regularization with DSP priors
- Stable and scalable inference

3. State-space representation

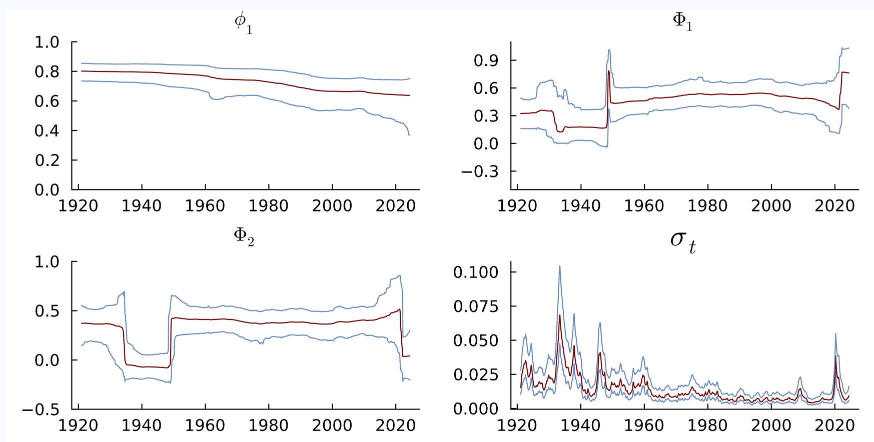
$$y_t = \boldsymbol{x}_{0t}^\top \tilde{\phi}_t^{(AR)} + \boldsymbol{z}_{0t}^\top \tilde{\psi}_t^{(MA)} + \varepsilon_t, \ \varepsilon_t \sim N(0, \sigma^2)$$

$$\tilde{\phi}_t, \tilde{\psi}_t = \tilde{\boldsymbol{g}}(\theta_t) \quad \leftarrow \textbf{stability/invertibility constraint}$$

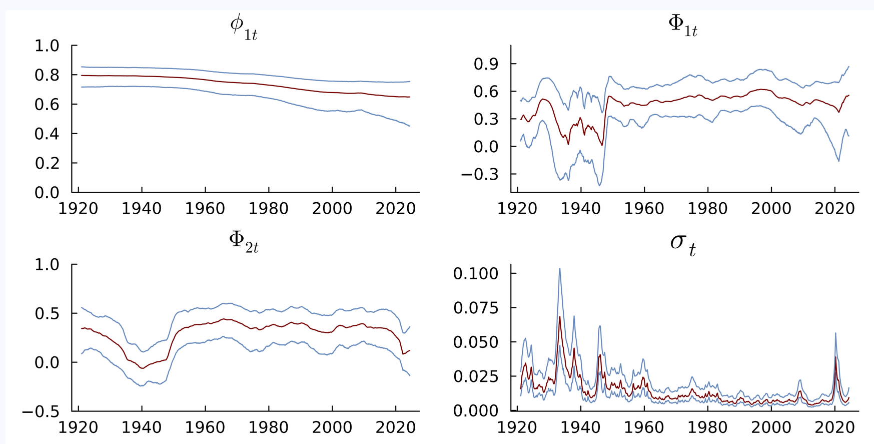
$$\theta_t = \theta_{t-1} + \nu_t, \qquad \nu_t \sim N(0, \mathrm{Diag}(\exp(\boldsymbol{h}_t))) \quad \leftarrow \textbf{DSP prior}$$

$$\boldsymbol{h}_t = \boldsymbol{\mu} + \boldsymbol{\kappa}(\boldsymbol{h}_{t-1} - \boldsymbol{\mu}) + \boldsymbol{\eta}_t, \qquad \eta_t \sim Z(1/2, 1/2, 0, 1).$$

4. DSP vs. Gaussian prior

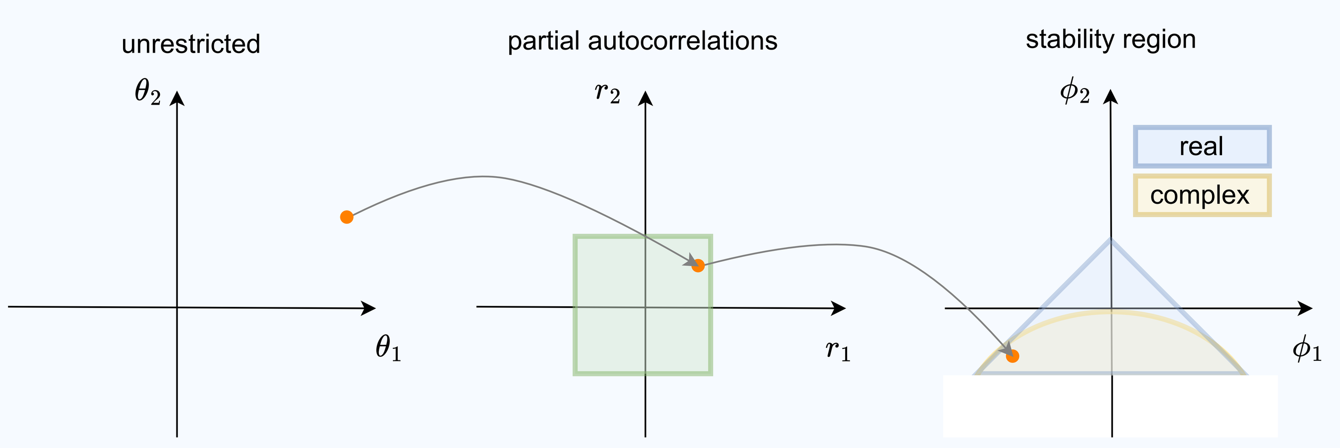


DSP prior with SV



Gaussian prior with SV

5. Stability parametrization and uniform prior

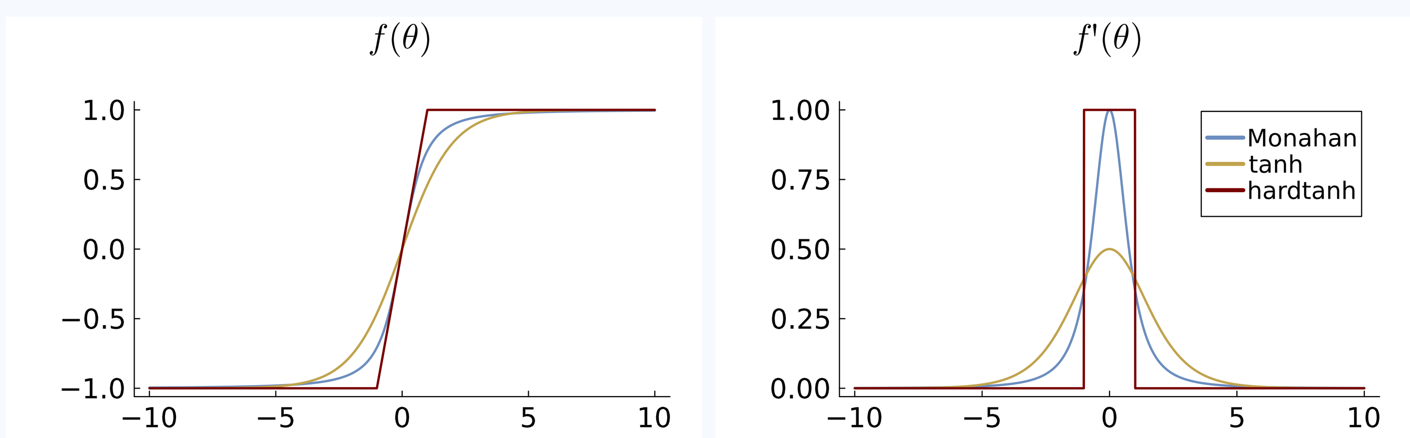


Monahan transformation + uniform prior over AR parameter space

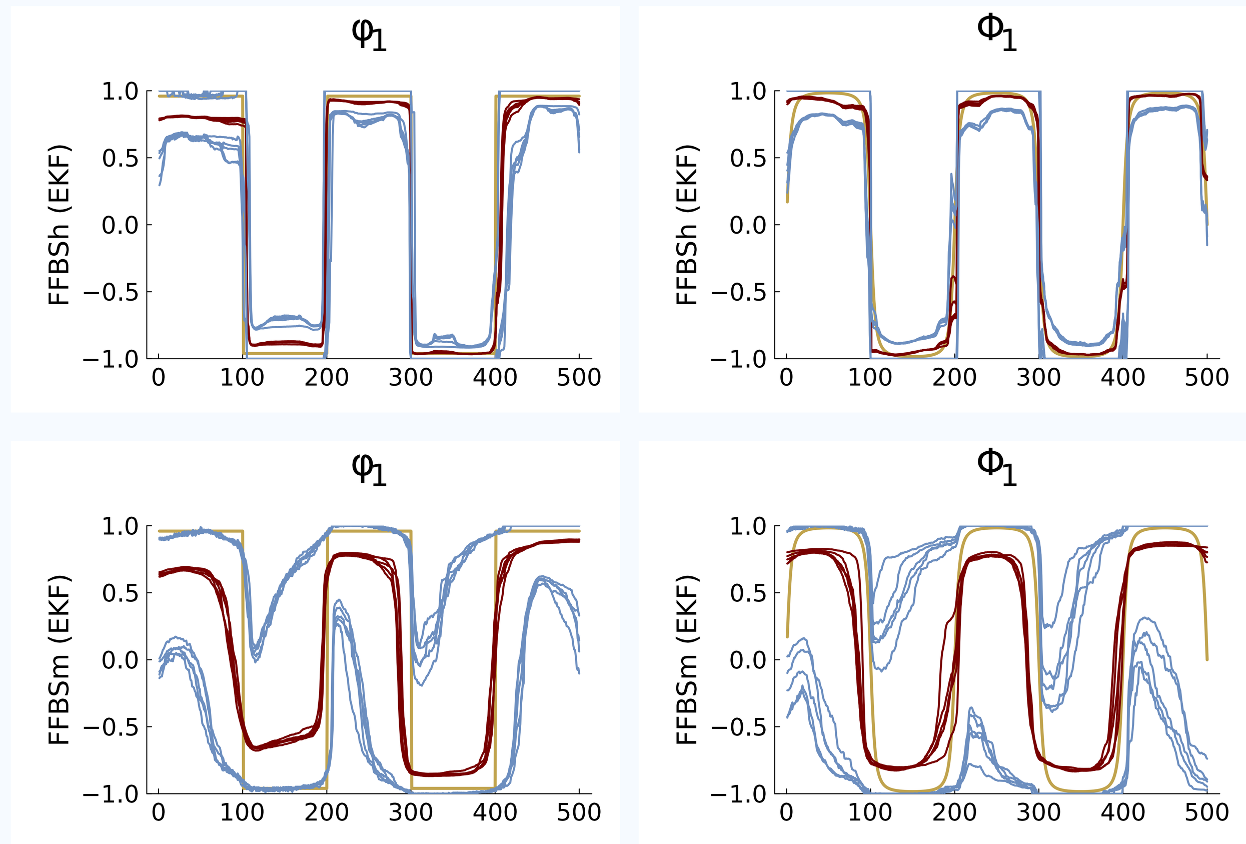
(Same idea applies to ensure invertibility!)

6. Stability parametrization at the boundary

Monahan (1984) $\Longrightarrow$ tanh $\Longrightarrow$ **hardtanh**



7. Parametrization effect on inference

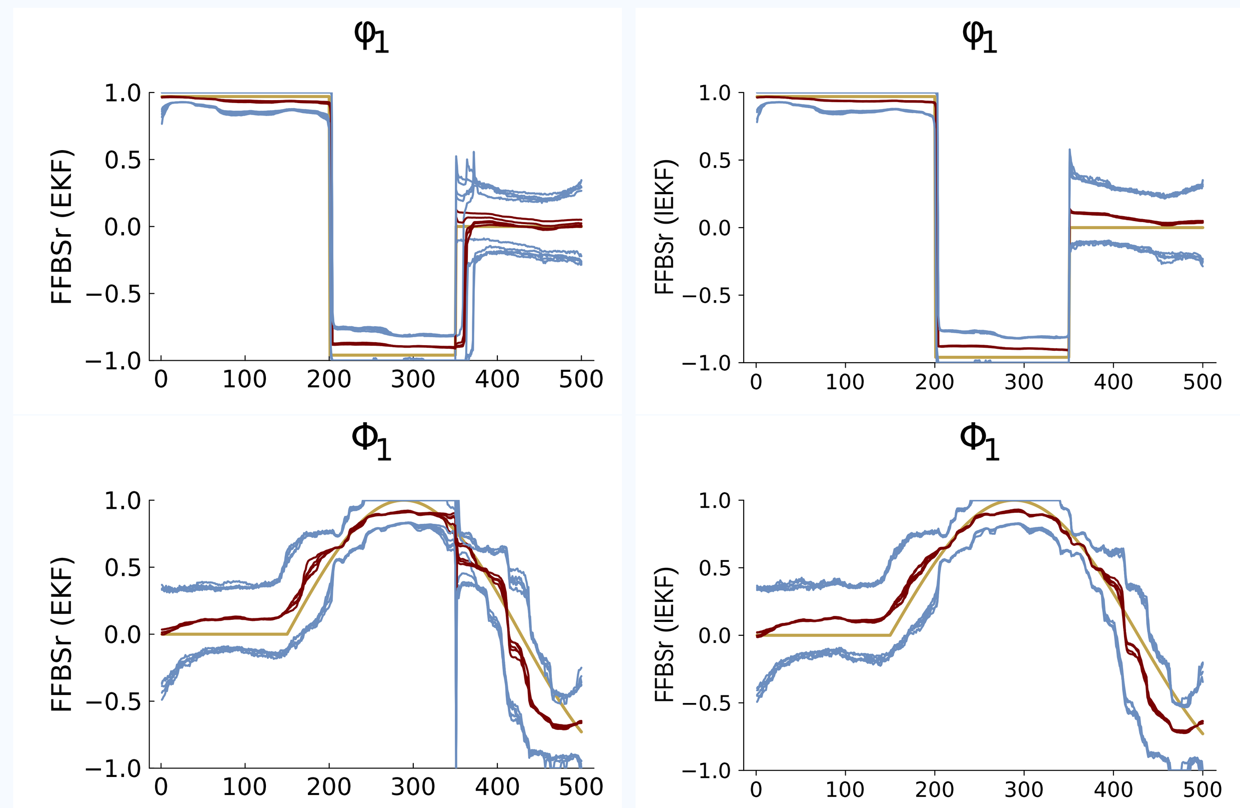


TV-SAR$(1,1)_{(12)}$: **Hardtanh** (first row) and **Monahan** (second row)

(Extreme example!)

8. Adaptive inference

FFBS+EKF (*Baseline*) $\xrightarrow{\text{NIS trigger}}$ **FFBS+IEKF** (*Adaptive*)



TV-SAR$(1,1)_{(12)}$: Baseline (first column) and Adaptive (second coulmn)

Fast when possible, accurate when necessary!

9. Gibbs sampling scheme

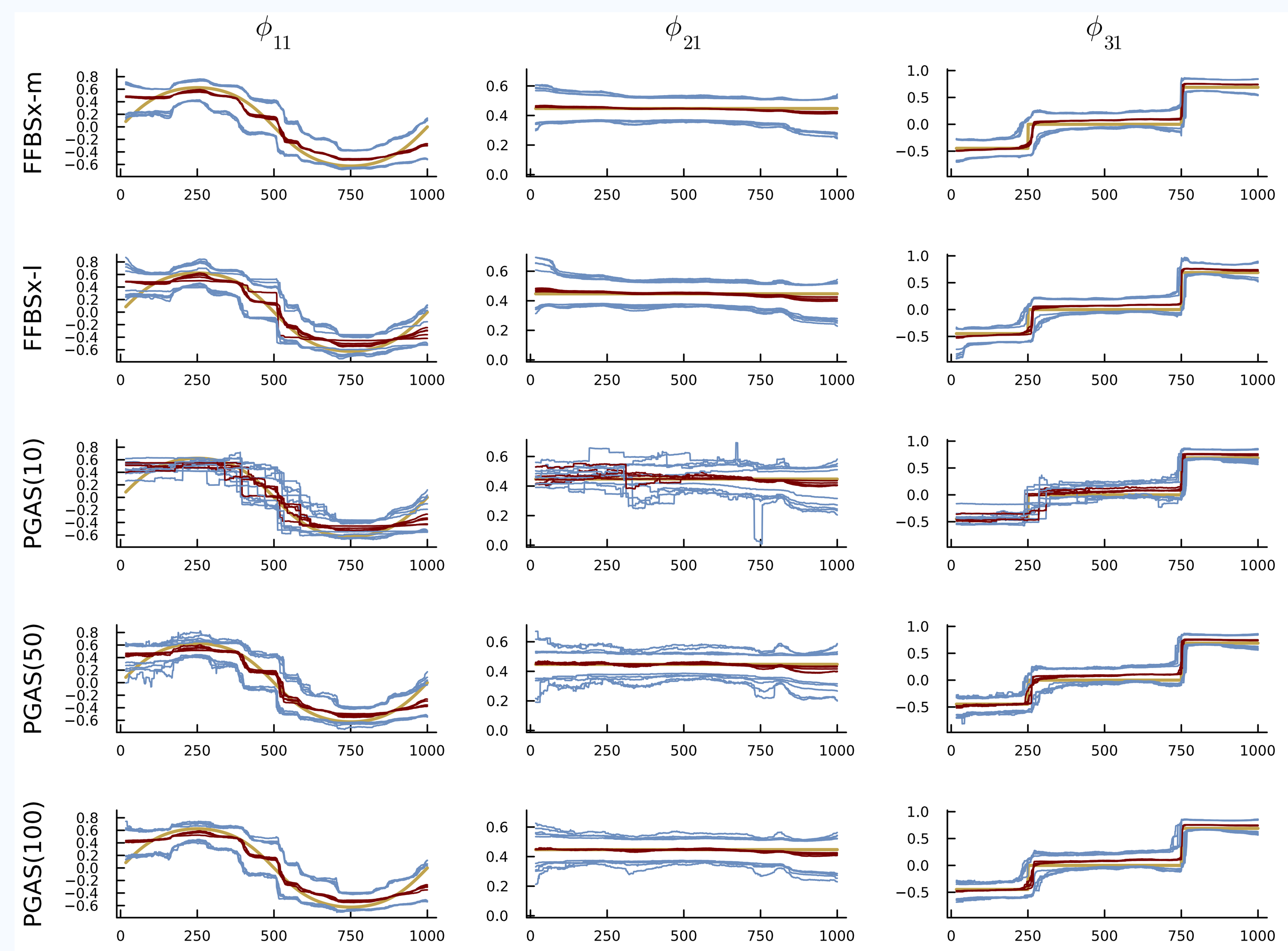
States $\rightarrow$ **DSP** $\rightarrow$ **ARMA** $\rightarrow$ **Presample** $\rightarrow$ **Errors** $\rightarrow$ **Volatility**

10. Scaling to multiple seasonalities

Grouping $\Longrightarrow$ **Sequential EKF** $\Longrightarrow$ **Scalable Inference**

(*Fewer state updates*) (*Lower filtering cost*) (*Long high-frequency series*)

11. Simulations



TV-SAR$(1,1,1)_{(4,12)}$: Posterior median (red) and 95% credible bands (blue)

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